

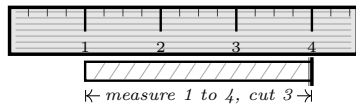
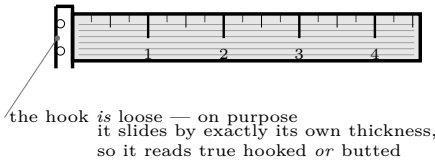
Craft 1: Measure & Mark

Everything downstream inherits the error you make here

Shop skill — transferable to anything you
ever build

THE STANDARD

A mark is **one line**, thin enough that you can tell which side of it to cut on, drawn square around the whole circumference of round stock, and made from a measurement you took twice. If you cannot say which side of your own mark the material comes off, the mark is not good enough.



The burnt inch. Start at a whole number, not the hook.
Subtract it. A worn hook then cannot lie to you.

Measure twice

Not a proverb — a procedure. Take the measurement, put the tape down, do something else for ten seconds, take it again. If the two disagree, take a third. Two agreeing measurements is evidence; one is a hope.

Mark thin

A pencil line is about half a millimetre wide, which on four joints is two millimetres of accumulated slop. Use a fine mechanical pencil, or scribe with a knife where a scratch is acceptable.

Mark which side

Put a small **X** on the waste side of every line, immediately. Cut so the blade eats the X and the line survives. This single habit removes most short-piece mistakes.

Go all the way round

On pipe, a mark on one side tells you nothing about the far side. Roll the pipe against a square, or wrap paper round it, and carry the line the whole way.

One reference Measure every cut from the *same* end of the stock, not from the last cut. Measuring cut-to-cut adds every previous error to the next one.

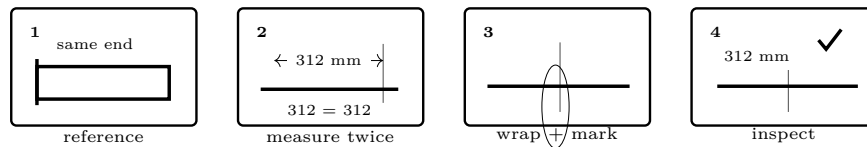
Write it down Cut list first, on paper, before any tool comes out. see the **Build Log sheet**

The wrap trick, and why it works

Wrap a sheet of paper around the pipe so the edge meets itself exactly. Because the paper edge is straight and the pipe is round, the line where the paper overlaps is automatically perpendicular to the pipe axis. Draw along it.

You should see: The line closes on itself when you have gone all the way round. If the two ends of the line miss each other, the paper was not aligned and the mark is not square — start again rather than splitting the difference.

Proof before the tool starts



1. Choose one clean reference end. Ask: *Is the hook seated, the tape straight, and my eye directly above the graduation?*
2. Measure, put the tape down, pause, and measure again from that same end. Write both readings. They must agree within 1 mm.
3. Wrap the paper with its edge exactly meeting itself. Draw one thin line around the pipe and put an X on the waste side.
4. Remove the paper. Ask: *Does the line meet itself? Is the X on the side I intend to discard? Does a final measurement still match the cut list?*

Pass, correct, or remake

Pass: two measurements agree within 1 mm, the line closes within 1 mm, and the waste side is unambiguous. **Correct:** erase a faint, wrong mark completely and repeat the whole sequence. **Remake the cut plan:** if several dark lines cannot be distinguished, move to sound unmarked stock; never guess which line was intended.

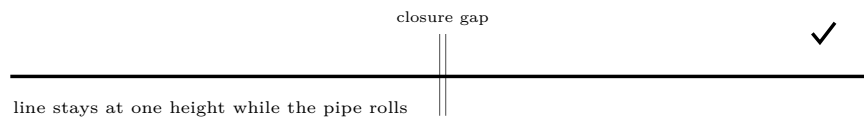
Where error comes from, in order of size

Source	Typical	What to do
Measuring cut-to-cut	several mm	Always measure from one reference end.
Thick pencil line	0.5 mm each	Fine pencil or knife; cut to one consistent side.
Tape not flat or not straight	1–2 mm	Pull it taut against the work, sight down it.
Reading the wrong graduation	1.5 mm or more	Say the number out loud as you read it.
Worn or bent hook	up to 1 mm	Use the burnt inch and it cannot matter.
Parallax — eye off to one side	under 1 mm	Look straight down at the mark.

Measurement and closure record

Use one row for each intended cut. Keep the tape seated at the same reference end for all three readings. Measure the closure gap where the two ends of the wrapped line meet; write 0 if they meet exactly.

Cut	First (mm)	Second (mm)	Final (mm)	Closure gap (mm)	Pass / correct
1					
2					
3					
4					



Check it yourself. Record the proof: first reading _____ mm; second reading _____ mm; final reading _____ mm. Roll the pipe slowly: does the line stay at one height? _____. Does the line close within 1 mm? _____. Is the waste X visible? _____. All answers must pass before cutting.

Why this sheet is first

Every later skill is downstream of this one. A cut cannot be squarer than its mark; a joint cannot be tighter than its cut; a chamber cannot be stronger than its joint. Accuracy is not a talent you have or lack — it is a set of small habits, and they are all on this page.

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